

Teorija polj

Vektorski operatorji

f = skalarno polje, F = vektorsko polje

grad: $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$

div: $\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

rot: $\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = (Py - Qz, Pz - Rx, Qx - Py)$

Laplacijan: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$

Lastnosti polj

Irotacijalno: $\nabla \times \mathbf{F} = 0$ Solenoidno: $\nabla \cdot \mathbf{F} = 0$

Potencial: $\vec{v}(x, y, z) = \textit{gradu}(x, y, z)$

u = skalarno p. in potencial polja v

Prostorska geometrija

Koordinatni sistemi

Polarni: $x = r \cos \theta, \quad y = r \sin \theta \quad J = r$

Cilindrični: $x = r \cos \theta, \quad y = r \sin \theta, \quad z = z \quad J = r$

Sferični: $x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta,$
 $z = r \cos \phi \quad J = r^2 \sin \phi$

Parameterizacija, odvodi integrala, tangenta, kot med vektorjema

Ločna dolžina: $s = \int_a^b |\mathbf{r}'(t)| \, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} \, dt$

parame. točk: $\vec{r} = r_A * t(r_B - r_A)$

Odvod integrala s parametrom:

$$F(x) = \int_{u(x)}^{v(x)} f(x, y) dy \quad \Rightarrow$$

$$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial(f(x, y))}{\partial(x)} dy + f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x)$$

Naravna parameterizacija: $|\mathbf{r}'(s)|^2 = 1, \quad s = \text{ločna dolžina}$

Tangentna premica in normalna ravnina: $\vec{e}$ ali $\vec{n} = \dot{\vec{r}}(t_0)$

Kot med Vektorjema: $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

Večkratni integrali

Dvojni integral: $\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$

Ploščina: $A = \iint_D 1 \, dA$

Prostornina: $V = \iiint_D f(x, y) \, dA$

Težišče: $\bar{x} = \frac{1}{A} \iint_D x \, dA, \quad \bar{y} = \frac{1}{A} \iint_D y \, dA$

Trojni integral: $\iiint_E f(x, y, z) \, dV$

Masa: $m = \iiint_E \rho(x, y, z) \, dV$

Vztrajnostni moment: $I = \iiint_E r^2 \rho \, dV$

Povprečna vrednost: $\bar{f} = \frac{1}{V} \iiint_E f(x, y, z) \, dV$

Krivuljni integrali

S. polje: $\int_C f(x, y, z) ds = \int_{t_0}^{t_1} f(x(t), y(t), z(t)) * \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt$

V. polje: $\int_C \mathbf{V} \cdot d\mathbf{r} = \int_C (Pdx + Qdy + Rdz) =$
 $= \int_{t_0}^{t_1} (P\dot{x} + Q\dot{y} + R\dot{z}) dt \quad d\vec{r} = \dot{\vec{r}} dt = (\dot{x}dt, \dot{y}dt, \dot{z}dt)$

Neodvisen od poti če $\vec{v}$ potencialno polje ($\textit{rot}\vec{V} = 0$)

Ploskovni integrali

S. polje: $\iint_S f \, dS = \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2} \, du \, dv$

kjer $E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$
 $z = z(x, y), \quad \vec{n} = (-z_x, -z_y, 1)$

V. polje: $\iint_S \mathbf{V} \cdot d\mathbf{S} = \iint_S \mathbf{V} \cdot \vec{v} \, dS = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v] \, du \, dv$

$$\vec{v} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}, \quad |\vec{r}_u \times \vec{r}_v| = \vec{n}$$

Integralski izreki

Greenov: $\oint_C (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$

C : zaključena pozitivno orientirana krivulja, ki je rob območja D

Stokesov: $\oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{V}) \cdot d\mathbf{S}$

C : rob površine S

Gaussov: $\iiint_S \mathbf{V} \cdot d\mathbf{S} = \iiint_E \nabla \cdot \mathbf{V} \, dV$

S : zaprt površina, ki je rob območja E

Kompleksna analiza

Elem. komp. fun.

$$z = x + iy, \quad |z| = \sqrt{x^2 + y^2}, \quad \arg z = \arctan\left(\frac{y}{x}\right)$$

$$e^z = e^x (\cos y + i \sin y)$$

$$\log z = \ln |z| + i(\arg z + 2\pi k)$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2}$$

Analitičnost in odvedljivost in harmoničnost

C-R pogoj: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Če velja je fun. analitična in odvedljiva

Pogoj harmoničnosti: $\nabla^2 u = \nabla^2 v = 0 / w_{xx} + w_{yy} = 0$

Formule kompleksne analize

Cauchy's formula: $f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$

Residue theorem: $\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k)$

$$\textit{Res} f(z) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]$$

Laurent series: $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$

Konformne presl. (C-R), Lomljena Linearna presl. / Mobiusova presl.

Preslikava: $w = f(z) = u(x, y) + iv(x, y)$

$$f(z) = \frac{az + b}{cz + d}; ad - bc \neq 0$$

Integrali

Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x)$

Po delih: $\int u \, dv = uv - \int v \, du$

Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}}$

Elementarni integrali

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

Odvodi

Pravila odvajanja

$$(f + g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Elementarni odvodi

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

Trigonometrijske funkcije

$$\sin^2 x + \cos^2 x = 1$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\sin 2x = 2 \sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$